Self-regulating accessibility: the unfrozen $d2{+}d4$ network never enters the compositional phase

interpolation from frozen-$d2$ to fully simultaneous training — 7 pretraining durations × 8 global λ × 3 seeds

Design

Same 4-bit, full-cube, two-block ReLU residual setup as the frozen experiments (N=1024, init variance $1/N$, exact Walsh, full-batch SGD). **Phase 1** pretrains *everything* (embedding, both blocks, 3-output readout with the $d4$ head masked) on the parents x_1x_2, x_3x_4 with $\lambda = 0$, saving checkpoints (parameters *and* momentum buffers). **Phase 2** resumes from each checkpoint with $d4$ supervision on, the $d2$ losses kept, *all layers trainable*, and a global weight decay $\lambda \in \{0, 10^{-4}, 3{\cdot}10^{-4}, 10^{-3}, 3{\cdot}10^{-3}, 10^{-2}, 3{\cdot}10^{-2}, 10^{-1}\}$ on all four weight matrices; within a seed every λ starts from the identical training state. Circuits are classified from block-2 pre-activation Walsh shares as before.

A design fact discovered at launch: $R_0 = Q_2/Q_1$ **at the input to block 2 is not freely dialable when nothing is frozen**. Random ReLU features already give $R_0 \approx 0.19$ at initialization, and converged $d2$ pretraining saturates at $R_0 \approx 0.33$ — the $1{+}1 \to 2$ corner units emit singletons of the same per-neuron magnitude as their pair, so Q_1 grows in lockstep with Q_2, capping R_0 *below* $R_c = 1/\sqrt{8} \approx 0.354$ (with a brief overshoot to 0.34 at $t_{\text{pre}} \approx 40$). Pretraining *duration* ($t_{\text{pre}} \in \{0, 5, 15, 40, 100, 400, 10^4\}$, from fully simultaneous to fully converged) is therefore the controlled axis, and R_0 the measured covariate.

Findings

1. The direct circuit wins everywhere. Across all 168 cells the compositional fraction never exceeds $f_{2+2} = 0.084$, while $d4$ is always solved ($\min|\hat{f}_4| = 0.85$, no failures). The (R_0, λ) phase diagram (Fig. 1, left) is uniformly in the 1^4 phase.

2. Pretraining history washes out. At $\lambda = 0$, f_{2+2} moves only $0.044 \to 0.060$ between $t_{\text{pre}} = 0$ and $t_{\text{pre}} = 10^4$, although R_0 spans $0.19 \to 0.34$. All conditions converge to the *same* $R_{\text{final}} \approx 0.31$: whatever the starting accessibility, joint training drives the representation to one fixed point below R_c (Fig. 2 — trajectories from both directions meet in the same band, never approaching R_c).

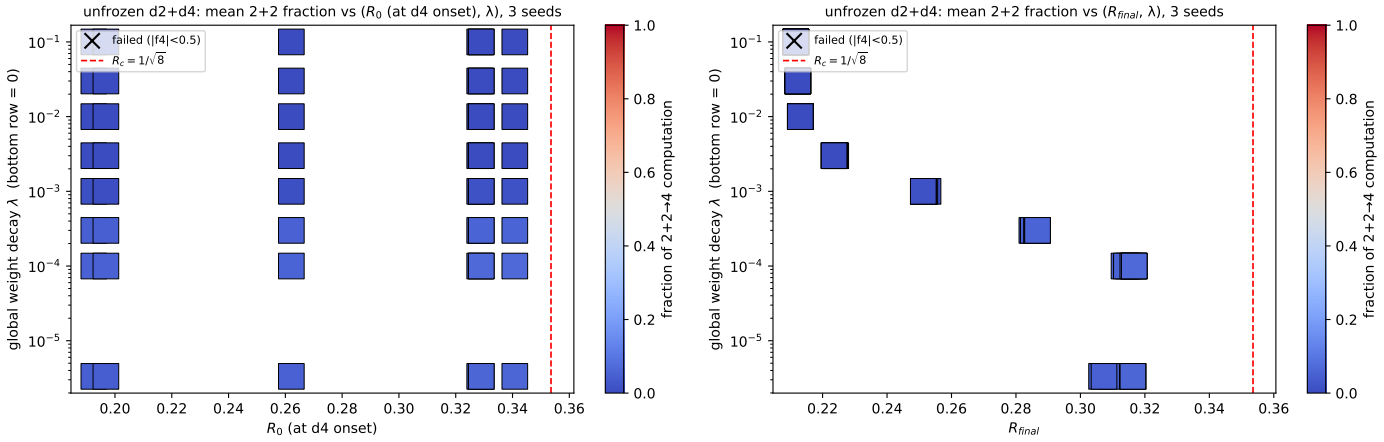


Figure 1: Phase diagrams of the unfrozen experiment vs. R_0 (left) and R_{final} (right); color = mean f_{2+2} on the same scale as the frozen-case diagram. Everything stays in the 1^4 (blue) phase, left of R_c (red). Note in the right panel how increasing λ *lowers* R_{final} while all pretraining durations collapse onto nearly the same point per λ.

3. Global decay lowers accessibility — the opposite of its frozen-case role. With λ acting on $V^{(1)}$ too, decay prunes the corner machinery to the minimum that still solves $d2$: $\|V^{(1)}\|_F$ falls $32 \to 1.8$ and R_{final} falls $0.32 \to 0.21$ as $\lambda: 0 \to 10^{-1}$ — an inverse α-rescaling. Consequently f_{2+2} hits exactly 0 by $\lambda = 3{\cdot}10^{-3}$. In the frozen setup λ sharpened the compositional phase right of R_c; here it destroys the conditions for it.

4. Frozen and unfrozen collapse against R_{final}, not against R_0. At matched $(R_{\text{final}}, \lambda)$ the two experiments agree quantitatively — e.g. at $\lambda = 0$, unfrozen $f_{2+2} = 0.060$ at $R_{\text{final}} = 0.32$ vs. frozen 0.052 at $R = 0.32$; at $\lambda \geq 3{\cdot}10^{-3}$ both give ≈ 0 for $R \leq 0.32$. Against R_0 they disagree strongly: frozen at $R = 0.34$, $\lambda = 3{\cdot}10^{-3}$ would give $f_{2+2} \approx 0.3$–0.4, the unfrozen network gives 0.000. In the spec's terms: **representation adaptation during $d4$ learning explains**

the difference — accessibility remains the governing variable, but it must be evaluated on the final, self-consistently adapted representation.

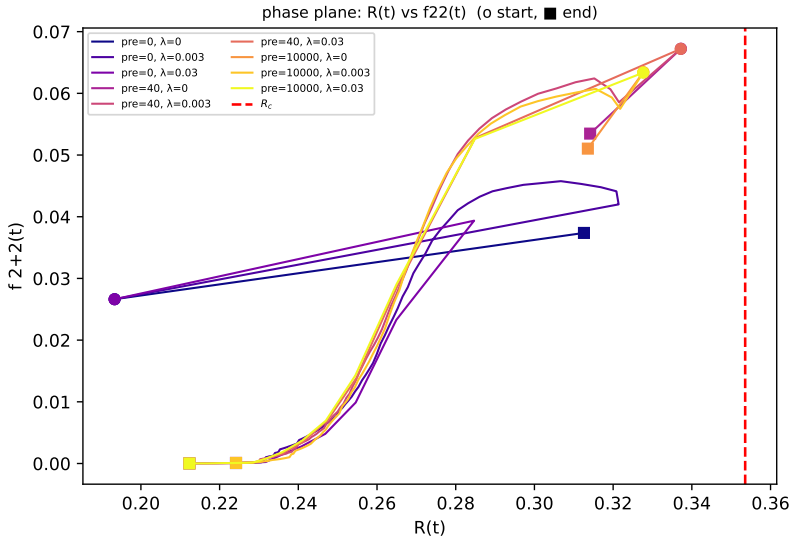


Figure 2: Phase-plane trajectories $R(t)$ vs. $f_{2+2}(t)$ for representative conditions (seed 0; ∘ start, ■ end). Cold colors start from scratch ($R_0 = 0.19$) and drift right; warm colors start pretrained ($R_0 \approx 0.33$) and, under decay, drift left — all end in the same sub-critical, low-f_{2+2} band. $R(t)$ never crosses R_c.

Conclusion. In this task the compositional 2+2 phase is reachable only with an externally pinned (frozen or rescaled) representation. Left to its own dynamics the network self-regulates to $R < R_c$ — corner units cannot help emitting the singletons that dilute their own pair advantage — and weight decay, applied globally, reinforces the direct route by shrinking the parent amplitudes themselves. A compositional phase under joint training would require either parents whose production does not carry singleton by-products, deeper targets (where the direct route's $\eta \propto Q_1/2^p$ collapses faster), or decay that spares the representation-carrying layers.

Artifacts: `gdrive:parity/analysis_output/accessibility_unfrozen/` (`results.csv`, `trajectories.csv`, `pretrain.json`, figures incl. `trajectories_panel` this report); code: `analysis/accessibility/run_unfrozen_experiment.py`. Companions: `accessibility_report.pdf`, `accessibility_wd_report.pdf`.